

VMware Setup

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🔗 URL	https://ubuntu.com/download/server

VMware Environment & Network Setup — Complete Process

Objective

Set up an isolated 3-VM lab (Kali attacker, Ubuntu-Benign target, Ubuntu-Suricata sensor) on a shared virtual network, separate from the host's regular internet connection.

Step 1: Check Existing Network Configuration

Before creating new VMs, examined Kali's existing network setup to understand what was already available:

- Opened Kali's VM Settings → Network Adapter — found it had **two adapters already**: one on NAT, one on Custom (VMnet1, Host-only)
- Opened **Edit** → **Virtual Network Editor** in VMware Workstation to see existing virtual networks
- Found **VMnet1** already configured as Host-only with subnet **192.168.93.0/24**, DHCP enabled
- Found **VMnet8** as the default NAT network with a separate subnet (192.168.23.0/24)

Decision made: reuse the existing VMnet1 rather than creating a new dedicated VMnet, since it was already present and unused for anything else — simpler than creating fresh network infrastructure.

Step 2: Download Ubuntu Server ISO

Downloaded `ubuntu-26.04-live-server-amd64.iso` from Ubuntu's official releases page. Confirmed this was the correct choice (Server, not Desktop) since:

- Neither planned VM needed a graphical interface for its core role
 - Server edition has a much smaller footprint (disk, RAM) than Desktop
 - More realistic to real-world deployments, where servers/sensors are typically headless
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Step 3: Create Ubuntu-Benign VM

1. **File** → **New Virtual Machine** → Typical → Next
2. Pointed to the downloaded ISO
3. Named the VM: `Ubuntu-Benign`
4. Set location to `F:\CEH\Ubuntu-Benign` (avoided default OneDrive-synced path to prevent unnecessary syncing overhead and because C: drive had limited space)
5. Disk size: 20 GB, stored as a single file
6. Clicked **Customize Hardware** before finishing:
 - Memory: 2048 MB
 - Processors: 1-2 cores
 - **Network Adapter** (first, auto-added): set to **Custom** → **VMnet1 (Host-only)**
 - Clicked **Add** → Network Adapter → set second adapter to **NAT**
 - Removed an accidentally duplicated CD/DVD drive entry
7. Clicked **Finish** to create and boot into the installer

Installer walkthrough (Subiquity text-based installer)

- Language: default (English)
- Keyboard: default

- **Network configuration:** confirmed both interfaces auto-obtained DHCP addresses (one ~192.168.23.x from NAT, one ~192.168.93.x from VMnet1) — continued
 - Proxy: left blank
 - Mirror: left default
 - **Storage configuration:** selected "Use entire disk" with default LVM partitioning (accepted the default 10GB root allocation out of the 20GB disk, rather than manually resizing, since 10GB was plenty for this VM's simple role)
 - **Profile setup:** entered name, server name (`ubuntubenign`), username (`ub`), and password
 - Ubuntu Pro: skipped
 - **SSH Setup:** checked "Install OpenSSH server" — critical, since this enabled remote command-line access via SSH instead of relying on the clunky VMware console window for all future work
 - Featured Server Snaps: left everything unchecked
 - Installation ran to completion, selected "Reboot Now," pressed Enter when prompted to remove the installation medium
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Step 4: Create Ubuntu-Suricata VM

Repeated the same process as Step 3, with these differences:

- Name: `Ubuntu-Suricata`
 - Location: `F:\CEH\Ubuntu-Suricata`
 - Disk size: 25 GB (slightly larger, anticipating Suricata logs/rules and a later-added desktop environment)
 - Memory: 4096 MB (more than Benign, anticipating running Suricata + eventually a GUI + Wireshark)
 - Same two network adapters (VMnet1 + NAT)
 - Same installer steps: server name `ubuntusur`, username `us`, OpenSSH server enabled
-

Step 5: First Login and Network Verification (both VMs)

Logged into each VM at the console, then ran:

bash

```
ip a
```

Why: confirmed each VM received two IPs as expected — one from the NAT adapter (192.168.23.x range) and one from the VMnet1 adapter (192.168.93.x range). Recorded the VMnet1 addresses specifically, since that's the address other lab machines would use to reach each VM:

- Ubuntu-Benign → 192.168.93.139
- Ubuntu-Suricata → 192.168.93.140

Step 6: Install VMware Tools Equivalent and Update System (Ubuntu-Benign)

bash

```
sudo apt update && sudo apt upgrade -y
```

Why: applied all pending security/package updates before installing anything else, ensuring a current, patched baseline.

bash

```
sudo apt install open-vm-tools -y
```

Why: this is the Linux equivalent of "VMware Tools," intended to improve integration between host and guest (clipboard sharing, better display handling, etc.). In practice, clipboard sharing didn't work reliably on this headless Server edition VM — SSH from Kali became the actual practical solution for comfortably typing/pasting commands going forward.

Step 7: Install Target Services (Ubuntu-Benign)

bash

```
sudo apt install apache2 vsftpd -y
```

Why: Apache (port 80, web server) and vsftpd (port 21, FTP server) give this VM realistic services to represent a genuine attack target — something with open ports and running software for Kali to scan, probe, and attack, and for Suricata to observe and generate alerts against.

bash

```
sudo systemctl status apache2 sudo systemctl status vsftpd
```

Why: confirmed both services were reported as "active (running)," not just installed but idle/failed.

bash

```
curl http://localhost
```

Why: sent a request to the VM's own Apache server from within itself, confirming it actually serves a valid webpage response (returned the default Apache HTML welcome page source).

Step 8: Discover and Fix Kali's Network Conflict

When attempting to connect from Kali to the two new Ubuntu VMs, ping and SSH both failed. Investigated:

bash

```
ip a
```

(on Kali) — revealed `eth1` had IP `150.1.7.101/24`, **not** an address in the `192.168.93.x` range like the other two VMs. This was identified as a **leftover static IP configuration** from an earlier, unrelated exercise (likely prior EVE-NG networking practice), manually assigned rather than obtained via DHCP.

bash

```
nmcli connection show
```

Why: listed NetworkManager's connection profiles to identify exactly which one controlled `eth1` — found it was named **"Wired connection 2."**

bash

```
nmcli connection modify "Wired connection 2" ipv4.method auto  
nmcli connection modify "Wired connection 2" ipv4.addresses ""  
nmcli connection modify "Wired connection 2" ipv4.gateway ""
```

Why: reverted the connection profile from a manually-set static IP/gateway back to automatic DHCP mode, clearing the old hardcoded values so the interface could request a fresh, correct address.

bash

```
nmcli connection down "Wired connection 2"  
nmcli connection up "Wired connection 2"
```

Why: forced the connection to disconnect and reconnect, applying the new DHCP setting immediately rather than waiting for it to happen on its own.

bash

```
ip a
```

Why: confirmed `eth1` now correctly showed an address in the `192.168.93.x` range (specifically `192.168.93.130`), matching the subnet used by both Ubuntu VMs.

Step 9: Verify Full Connectivity Across All Three VMs

From Kali:

bash

```
ping -c4 192.168.93.140  
ping -c4 192.168.93.139
```

Why: sent 4 ICMP echo requests each to Ubuntu-Suricata and Ubuntu-Benign — both returned successful replies with 0% packet loss, confirming Kali could now reach both machines across the shared VMnet1 lab network.

Also tested SSH access directly:

bash

```
ssh us@192.168.93.140
```

Why: confirmed not just network-layer connectivity (ping) but also that SSH itself was reachable and authenticating correctly — the actual working method used for all subsequent command-line work on the Suricata VM.

Outcome

By the end of this phase, all three VMs (Kali, Ubuntu-Benign, Ubuntu-Suricata) could reach each other over the isolated `192.168.93.0/24` VMnet1 network, each also retained separate internet access via their individual NAT adapters, and Ubuntu-Benign was running realistic target services (Apache, vsftpd) ready to be scanned and attacked in the next phase of the project.